

PowerRaft: Portable and Amphibious Solar Electric Generator

Prepared for Investment Pitch (Dec. 8th)

Prepared by Team 3 Solar

Obi Page

Sean Kerr

Calvin Bramble

Lucas Miller

Albert Robu

December 8th, 2023

Executive Summary

Portable. Floating. Solar. - The PowerRaft mantra

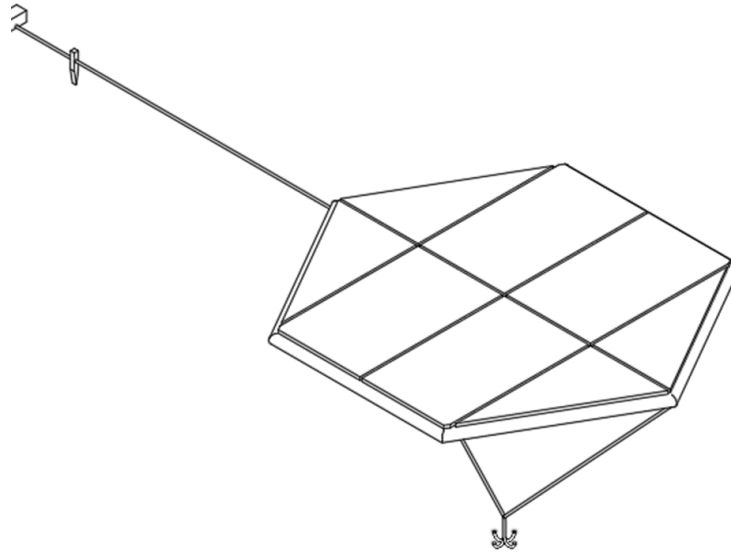


Figure 1: PowerRaft concept render [1]

This report is a detailed outline and description of the PowerRaft product. It explains both its premise and function, as well as presenting the case for why it is a worthy investment opportunity. Through a combination of good timing industrially and socially, as well as a truly novel design, the PowerRaft is primed for success. The PowerRaft is an expandable solar electric generating platform that consists of two main components: the serially connected solar array and the inflatable raft platform. The novelty of the device comes from its fundamental design concepts or pillars.

1. Foldability
2. Floatability

The first novel pillar of the design is its ability to fold. This allows the PowerRaft to be condensed for transportation and storage while being unfolded into a dramatically larger sun-exposed area. This allows the PowerRaft to be used anywhere it can be carried, which is significantly easier than with traditional solar panels of comparable size. This aspect opens up a world of remote and temporary applications for the PowerRaft. This foldability feat is accomplished with strategic seam design and adequate material selection for the inflatable. Additional factors include strategic device geometry as well as making the attached solar panel array (consisting of high efficiency solar “tiles”)

modular and removable. The 1000 watt concept PowerRaft can be condensed from a generating surface area of 5 square meters, to a size fit for a large backpack. This massive expandability aspect allows the PowerRaft to be far more powerful than any other portable or foldable solar panels currently on the market.

The second novel design pillar of the PowerRaft is that it can float. This puts it in line with a concept that is gaining tremendous global and North American momentum in the solar power and clean energy industries. The ability for the PowerRaft to be deployed on water, as well as land, allows it to be used in virtually any environment or situation. The floatability feature also allows it to be used out of the way of other activities. This reduction of land occupancy is the primary benefit of using floating solar, along with accessing otherwise obstructed sunlight (such as in a lakeside forest). With the floatability feature comes several other critical design components. These include the raft platform inflation process, the anchor system, as well as a tethering and power delivery system. This second novelty of the design separates it from any existing products, making it compatible with a variety of recreational applications like camping and boating.

The general strategy for the PowerRaft is two pronged.

1. Humanitarian Aid
2. Recreational Use

In both cases the PowerRaft will use its unparalleled portability and versatility to provide power anywhere at a practical scale.

Considering the global interest in renewable energy and the growth of the floating solar industry, the PowerRaft is a very likely success. This makes it a highly unique and lucrative investment opportunity that should not be ignored. It has the potential to greatly support infrastructure in humanitarian and disaster situations (providing usable levels of electrical power for communication and medical use). It also has strong revenue prospects as an aid for remote and aquatic recreation.

1. Introduction

The floating solar power industry is on the rise and is currently in the midst of massive growth globally and in North America. This growth and interest in the industry is a wave that the PowerRaft will capitalize on, filling the niche of a portable and prefabricated version. The introduction of this report provides more details on this current market state

as well as its historical context. In addition it will describe the history of the company (Team 3 - Habileos Solar) and the PowerRaft product.

1.1: Purpose of this Report

The purpose of this report is to provide information on the PowerRaft device, its intended use and the relevant market conditions. The aim is to solicit investor interest and funding.

1.2. History of Team 3 Solar

Team 3 Solar was founded in September 2023, with the aim of designing unique, solar based solutions for remote logistical challenges. Particularly, Team 3 Solar wanted to provide devices and products to improve lives both domestically and internationally and help the world achieve greater energy independence. Team 3 Solar aims to electrify the world independently from traditional, large footprint grid infrastructure and provide equal access to locally generated, renewable electricity. The PowerRaft device was conceived to provide the most power in the most places possible. Thanks to the work of the design team at Team 3 Solar, as well as investor funding, the PowerRaft vision is nearly a reality.

1.2.1: Current Team 3 Solar Members

Currently, Team 3 Solar consists of 5 design team employees.

- Obi Page - solar panel array design.
- Sean Kerr - Raft material and general PowerRaft device design.
- Calvin Bramble - raft buoyancy design.
- Lucas Miller - Anchor and power tether system design.
- Albert Robu - react inflation procedure design.

1.2.2: Inception of the PowerRaft

The PowerRaft was conceived in response to logistical challenges of accessing electricity in remote locations. From transporting temperature sensitive vaccines along the Amazon river, to powering campsites on Canadian lakes, the PowerRaft can be there. Portable and versatile, the PowerRaft was designed to electrify any emergency or recreational scenario.

1.3 Current Marketplace for Product

The market for Floating Photovoltaics (FPV) is experiencing significant growth due to a recognition of the benefits it offers compared to traditional solar power installations. Coupled with an increasing interest in renewable energy alternatives, the PowerRaft is a unique entry into the niche of portable FPV power generation.

1.3.1: Floating *Photovoltaics* (*Floatovoltaics*)

The FPV market is in a state of incredible growth that is continually accelerating. The global production of FPV power has increased from 1.1 megawatt hours (MWh) in 2009, to 117, 965 MWh in 2021 [2]. Additionally, in September 2023, Indonesia activated the world's largest floating solar power generation system in a reservoir in West Java. Capable of producing around 300 MWh annually, it is able to power over 50,000 homes year-round [3]. Despite historical popularity in Asia, the concept of FPV generation has been gaining traction in North America, where plans for new projects are increasing [4].

The PowerRaft is unique in that it provides a temporary installation generating option that is highly portable and also floats. This sets it apart from both traditional portable solar electric generating devices as well as more permanent floating solar systems.



Figure 2.1: The world's largest floating solar farm in West Java, Indonesia [5]

1.3.2: Benefits of Floating Photovoltaics

The primary benefit of FPV systems is that it reduces land occupancy. Depending on the situation, land may be limited or needed for other uses (such as farming or habitat preservation in the case of larger solar farms). By installing solar panels on a body of water, this eliminates the significant obstruction it would pose on land. Installing generating systems on water also allows for construction closer to urban areas that have extremely finite available land area [6]. By generating the power closer to where it will be used, it reduces the cost and environmental impact of necessary infrastructure. An additional benefit of installing solar panels on water is that the aquatic environment provides passive thermoregulation of the panels. This is noteworthy because solar

panels are known to decrease in efficiency when they exceed a temperature of 25°C [7]. Therefore, FPV systems can be (especially at larger scales) more efficient than traditional photovoltaic (PV) systems. These benefits are all factors as to why the FPV industry is gaining momentum.

1.3.3: Disaster Relief/Infrastructure support market

As mentioned in the executive summary of this report, one of the primary applications of the PowerRaft is for humanitarian aid. The PowerRaft product has the portability, versatility and scale of output to be a game changer in situations of recovery from natural and manmade disasters. The feasibility of portable PV systems in this context was investigated in a 2018 analysis of a portable photovoltaic system in Malaysia. That analysis showed that the system was able to meet 99% of the daily load demand year-round (excluding December) [8]. With a theoretical output of 1 kilowatt, the PowerRaft could contribute to a temporary microgrid, or as a standalone generator (with a battery system). It could be used to power various critical devices related to communication, medical and survival systems (such as radios, lights, refrigeration units, water pumps, etc.) The portability of the PowerRaft allows it to be deployed quickly, wherever it is needed. It would provide expedient relief for people in need of electrical power, quicker than a less portable and less self contained generating system.



Figure 2.2: Solar panels being prepped for transport on the Amazon river, Brazil [9].

1.3.3: Recreational Markets

As the main revenue stream, the use of the PowerRaft for recreational applications is critical for the device's financial success. According to the Kampgrounds of America, 58 million North American households reported camping at least once in the year 2022 [10]. This camping market is coupled with an increasing reliance on cell phone and portable electronic devices, as well as expanded cellular network coverage. This means

the desire for convenient power while camping is increasingly significant. Additionally, boating and fishing are the largest recreational activities in the United States, comprising an industry worth \$30.8 billion USD [11]. The combination of these and other recreational markets provide the PowerRaft with a significant potential customer base for revenue generation.

1.4: Historical Factors of the Marketplace

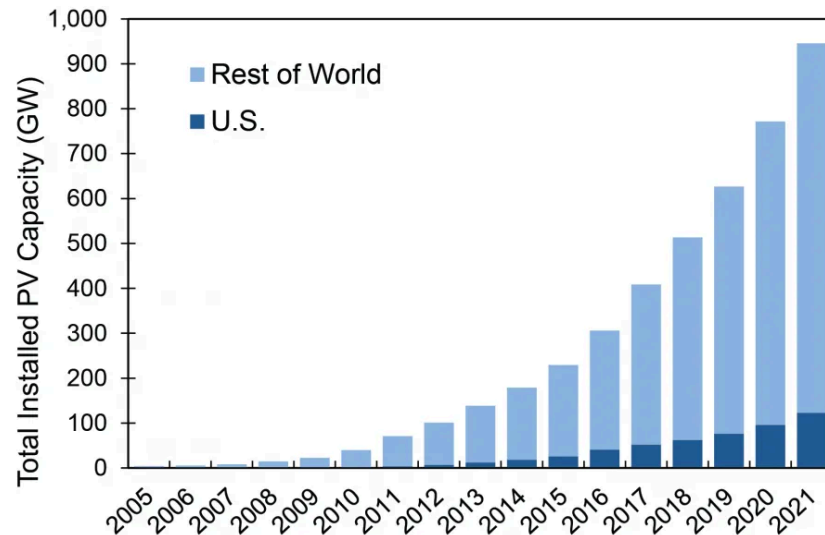


Figure 3.1: Increase in solar power production both globally and in the United States [12].

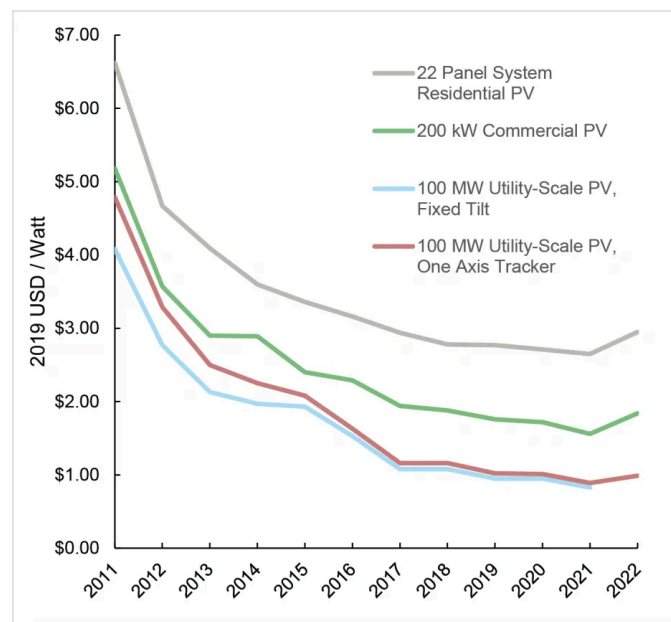


Figure 3.2: Decreasing cost of solar power production over time [12].

The cost of solar panels has been steadily decreasing, while the efficiency has been increasing. This has made the market healthier than ever and getting healthier. It is also more accessible to get into solar energy production than at any other time in history. These market conditions are exemplified by figures 3.1 and 3.2, the cost of solar power generation has dramatically decreased between the years 2011 and 2022. Additionally, the global and US production of solar power has increased by nearly one billion watts between 2005 and 2021 [12]. With these historical market conditions considered, the timing is extremely favorable for the introduction of a product like the PowerRaft.

1.5: How to use this report

To use this report, simply follow the headings to find any desired content. The document is structured to be intuitive in presenting the PowerRaft product and describing the investment opportunity. A table of contents is provided at the front to aid in section location.

2. The PowerRaft

The PowerRaft consists of a modular solar array of small, flexible solar tiles that cover the top surface of an inflatable raft platform. It includes an anchor and tether system that allow it to generate power on a body of water and deliver it back to the shore. It is also foldable and lightweight for compact transportation and storage.

2.1: PowerRaft Design Overview and Function

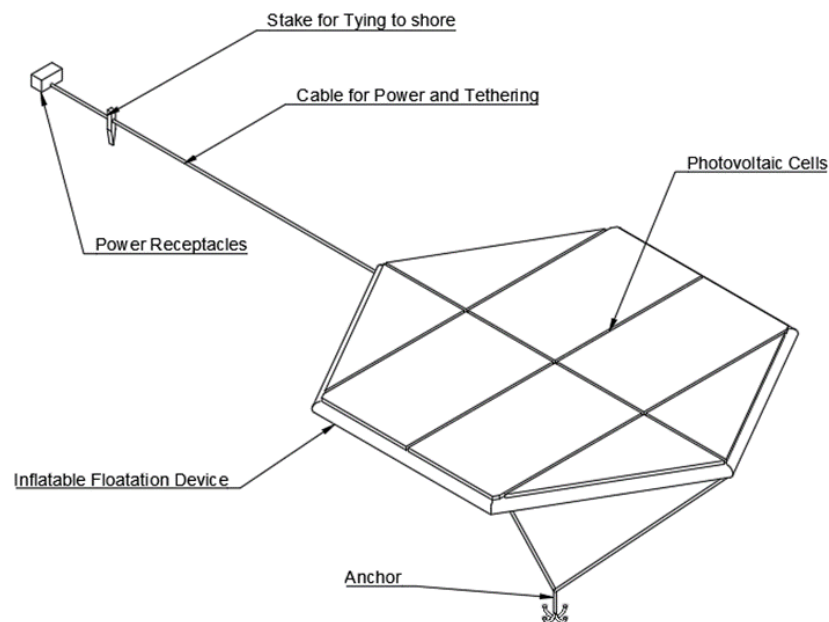


Figure 4: Labeled preliminary PowerRaft concept diagram [13].

The PowerRaft is designed with a specific geometry and seam placement for repeated folding and unfolding. From a hexagon with a sun facing area of 5 square meters, the raft can be condensed to a rectangular prism of roughly 0.6 x 0.7 x 0.07 meters. This coupled with the stack of lightweight mono-crystalline solar panel tiles will fit comfortably in a large backpack. At an efficiency of 20% for monocrystalline cells (some are 22+%) [14], a 5 m^2 generating area can output 1 kilowatt of power [15]. At these dimensions, with maximum solar panel coverage of the top side of the inflatable platform, the PowerRaft system will weigh approximately 38.1kg.

Table 1: 1kW concept model PowerRaft estimated preliminary specifications.

Maximum Power Output	Weight (Panel Array) [16]	Weight (Raft Inflatable) [17]	Raft Unfolded Sun Facing Area	Raft Folded Dimensions
≈ 1000 Watts (Joules/second)	14.1 kg	24.0 kg	5.00 m^2	0.600 x 0.694 x 0.072 m

The PowerRaft also includes a debris anchor (lightweight nets that hold rocks or other debris) and a power tether that leads to an output terminal. It is able to deliver power from a body of water to a connected battery system or alternative load. The solar panels will be flexible to conform to the surface of the raft platform and will be adhered with velcro.

2.2: PowerRaft Design Components

The PowerRaft consists of 2 primary design components.

- The photovoltaic/electrical system.
- The Raft Mounting Platform.

The raft platform itself then has several specific considerations.

- Material design.
- Buoyancy and floating stability design.
- Inflation procedure.
- Tether and anchor system.

Below is a detailed overview of each area of the PowerRaft design.

2.2.1: Photovoltaic/Electronic Design

The main operating principle of photovoltaic technology is the photovoltaic effect. This principle describes the phenomenon of electrons being dislodged from a conductive material by impacting photons (wave packets of light) [18]. In solar panels, this phenomenon is exploited by using semi conductive materials (silicon coated with a more conductive element) to generate an electron charge gradient. This charge gradient

spans the internal terminals of the panel, generating a current between them as the electrons move through to replace the electrons being dislodged.

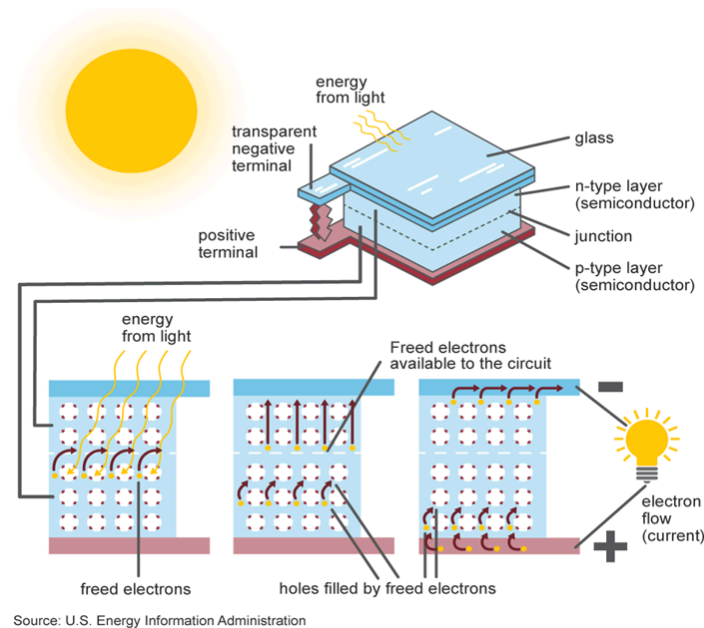


Figure 5.1: Diagram of the current generation inside a solar panel [19].

The solar electric aspect of the PowerRaft's design is paramount. To achieve maximum efficiency per square meter, monocrystalline solar cell panels have been chosen. However this efficiency involves increased manufacturing complexity raising the cost per panel (see figure 2.3 in the report introduction). Though, the price per watt of power production is continually decreasing for all cell types [20].

Additionally, the solar panels will be connected in series. This allows for a combination of each panel's voltage and power. This total power will be sent along the tether to an output transformer, where it will be converted for a 12V DC outlet terminal. This voltage was chosen for convenient compatibility with most portable DC electronic devices and battery systems.

2.2.3: Raft Material Design and Buoyancy:

The raft geometry has been designed as a hexagon for stability while afloat and space efficiency. The buoyancy of the raft is verified by the density of the system (weight divided by volume), relative to the density of water. The density of the 1000W concept PowerRaft when inflated is less than 3 kilograms per cubic meter, which is low enough for stable flotation.

The material chosen for the PowerRaft's inflatable raft component is Thermoplastic Polyurethane (TPU). The main principles explored while selecting the raft material were the specific strength, and the specific modulus. The specific strength of a material is the tensile load capacity relative to the density. This indicates the strength to weight ratio of a material, which is key for portability and durability. The specific modulus is the stiffness and flexibility of a material, relative to its density. In the case of the PowerRaft, this is critical for how the raft is folded for transport.

$$\text{Specific Strength} = \frac{\sigma_y}{\rho} \quad (1) \quad [21, \text{pp.50}]$$

$$\text{Specific Modulus} = \frac{E}{\rho} \quad (2) \quad [21, \text{pp. 50}]$$

σ_y = material strength, ρ = material density, E = material modulus of elasticity

TPU was chosen as the raft material for a variety of additional reasons, with the total criteria summarized below [22].

- High Tensile Strength (Specific Strength: $75 \frac{kN \cdot M}{kg}$)
- Highly elastic (Specific Modulus: $4.3 \frac{kN \cdot M}{kg}$)
- Extremely resistant to ultraviolet radiation damage.
- Abrasion resistant.
- Thermally Tolerant/Stable.

TPU is also fairly low-cost, at \$5.50 USD/kg [23]. At this price, the raft component of the 1000W model PowerRaft would cost around \$132 USD (see device overview for specifications).

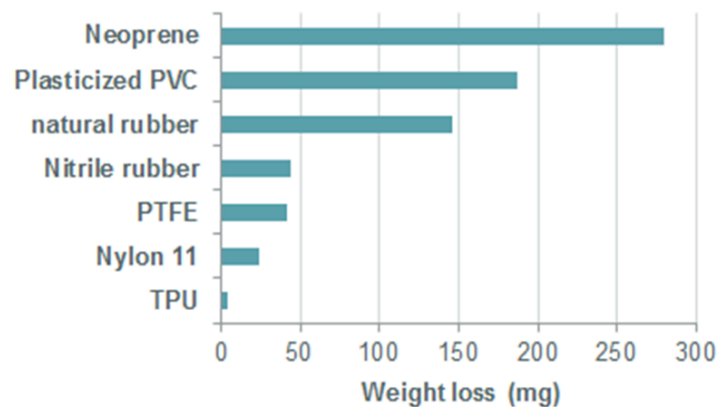


Figure 5.2: Abrasion Resistance (measured in material lost) of TPU compared with alternate materials [23].

2.2.3: Inflation Procedure

The raft will be inflated to a set internal pressure, monitored with a built in mechanical pressure gauge. This internal pressure will ensure maximum raft stability, as well as the optimum exposure angles for the attached solar array. A mechanical pump will be included with the PowerRaft for more convenient inflation.

2.2.5: Anchor and Tether System

The anchor system consists of two model versions. The first anchor system version consists of a collapsible boat anchor that is lowered and raised mechanically by a cable along the PowerRaft's tether. This version of the anchor provides increased stability to the PowerRaft while on the water, at the cost of reduced portability. The second version consists of a debris anchor (lightweight netting used to hold site acquired weights). The debris anchor alternative will also use the cable control mechanism. Both these anchor system types provide stability to the PowerRaft, and keep it secured to a set location during operation.

The PowerRaft tether consists of an insulated power cable surrounded by a ring of galvanized steel security cables. This allows the tether to both deliver power from the PowerRaft to the shore, while also securing it mechanically. The mechanical tether component also controls the raising and lowering of the anchor. A cross section of the cable is provided in figure 5.3 below.

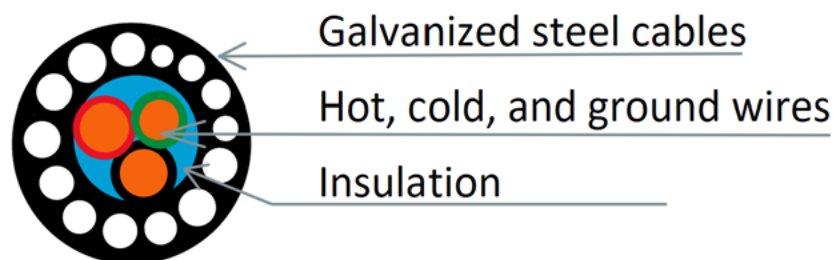


Figure 5.3: Cross sectional diagram of the PowerRaft tether [24].

3. Investment Value:

As outlined through both the PowerRaft design philosophy and specifications, as well as the conditions of the broader FPV industry and the targeted applications of the PowerRaft, there is potential revenue and impact potential for this device. It has the dual benefits of being highly novel, as well as genuinely practical, and is nearly assured in its high return on any potential investment.

3.1: Revenue Potential

The revenue potential of the PowerRaft comes from both the worldwide opportunities afforded by the healthy condition of the relevant industries, as well as the competitive advantage provided by the uniqueness of the PowerRaft's design.

3.1.1: Worldwide and Domestic Opportunities

The significant growth of the FPV industry provides a strong opportunity for all relevant technologies to gain interest and attention. Because the PowerRaft is distinct from the large scale floating solar installations, it exists independently in its own niche. This makes it more likely to be discovered and desired for recreational applications that had not previously been possible before its inception. The decrease in price and improvement in panel efficiency, and flexibility that has made the PowerRaft possible is an ongoing trend. This enables a future in which later iterations of the PowerRaft can be more powerful and lightweight. This potential for future improvement will allow the PowerRaft to appeal to even more customers as it becomes viable for even more applications.

3.1.2: Competitive Advantage

There is currently nothing on the market that does exactly what the PowerRaft does, nor as well. This dominance of the PowerRaft in the solar generator market is compounded by its versatility. It can go where no competitive solar generator can and it can stay out of the way, generating practical levels of power in the background. The niche that the PowerRaft occupies is exemplified in table 2.

Table 2: PowerRaft power output and weight compared to existing market alternatives.

	Solar Power Format Options				
	PowerRaft	Portable		Non-Portable	
Model	1000W Concept Model (Multiple Panels)	Renogy 175W Flexible Solar Panel [25]	Paso 100W CIGS Portable Solar Blanket [26]	JA Solar Jumbo JAM80S40 785-810 [27]	Tongwei G12 [28]
Maximum Power Output (W)	1000	175	100	810	780
Weight (kg)	38.1 (24.0 raft mount + 14.1 solar panels)	2.81 (panel only)	2.20 (folding panel mat)	43.5 (panel only)	39.0 (panel only)

The PowerRaft exists between the markets of portable and permanent installation panels, with a comparable weight and vastly increased portability to leading output non-portable panels. It also exceeds the power output of leading residential models, and with its self contained mounting structure, provides full scale power in a portable format.

3.2: Impact Potential

The impact potential of the PowerRaft is substantial, as it provides more power than any alternative renewable device in a more accessible format. While the recreational revenue will be sufficient in financing the production of the PowerRaft, the greatest impact potential for the device is in its humanitarian uses. An Engineers Without Borders report estimated that a 30,000 person refugee camp requires 600 kWh (600,000 watts for one hour) per day [29]. This energy demand is currently met with diesel generator systems, which have high operating costs due to the amount of fuel required. To reduce operating costs and bypass the logistical risks of fuel reliance, humanitarian agencies are increasingly looking to renewable and hybrid fossil fuel-renewable energy systems. The PowerRaft could be a key component in these future energy systems, providing clean power to remote locations in need at no recurring cost.

The PowerRaft also has the potential to help liberate people from municipal energy grids, letting them achieve clean energy independence. A future that consists of independent, renewable microgrids and distributed energy generation is far more sustainable: It reduces carbon emissions, increases access to electricity and provides greater electrical reliability for critical uses [30]. The PowerRaft has the potential for integration in any off grid electrical system, with its enhanced versatility putting it ahead of existing alternatives.

3.3: Report Conclusion

The information provided in this report shows that the PowerRaft is an important and timely product for today's energy market. From its various applications and revenue potential, to the profound novelty of this device, it stands apart from the competition as a near certain success. At an estimated price point of between \$1000-1500 for the entire 1000 watt PowerRaft system, it is cost competitive with less powerful models. In addition, as panel efficiency increases and costs decrease, this figure will come down significantly. As panel manufacturing and material science improves as well, the PowerRaft will get even more portable and lightweight. With all these factors considered, the PowerRaft is an important investment not to ignore.

On behalf of everyone at Team 3 Solar, thank you for reading this document and considering investing in the PowerRaft.

Team 3 Solar Contact Information:

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